

Traditional irrigation is more than just a production strategy

R code

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```

# PRELIMINARY FUNCTIONS #####

library(sensobol)
sensobol::load_packages(c("data.table", "ggplot2", "sf", "rnatuarearth",
                          "rnatuarearthdata", "ggrepel", "here", "readxl",
                          "ggwordcloud", "cowplot", "benchmarkme"))

# Source all .R files in the "functions" folder -----

r_functions <- list.files(path = here("functions"),
                          pattern = "\\..R$", full.names = TRUE)
invisible(lapply(r_functions, source))

```

1 Interview locations

```
# INTERVIEW LOCATIONS #####

locations <- data.table(
  place = c("Dúrcal", "Huétor-Tajar", "Murcia", "Ricote", "Sudanell", "Arriate", "Oria"),
  lon   = c(-3.5694, -4.0439, -1.1307, -1.4361, 0.6011, -5.0606, -2.2731),
  lat   = c(37.0022, 37.1964, 37.9922, 38.0714, 41.5261, 36.7914, 37.5225)
)

# SPAIN BOUNDARY #####

spain <- ne_countries(scale = "medium", country = "Spain", returnclass = "sf")
```

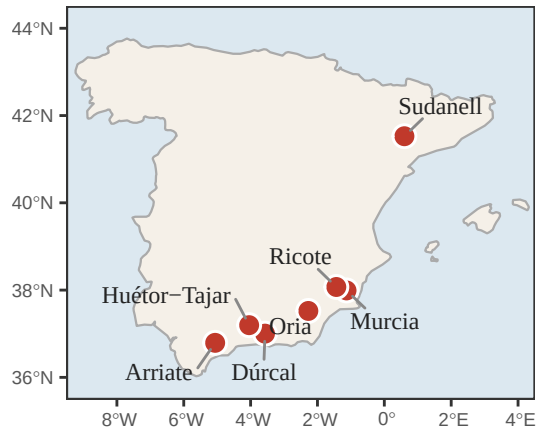
1.1 Plot map

```
# PLOT #####

plot_locations <- ggplot() +
  geom_sf(data = spain, fill = "#f5f0e8", color = "#aaaaaa", linewidth = 0.4) +
  geom_point(data = locations,
    aes(x = lon, y = lat),
    shape = 21, fill = "#c0392b", color = "white",
    size = 3.5, stroke = 0.8) +
  geom_text_repel(data = locations,
    aes(x = lon, y = lat, label = place),
    size = 3.2, family = "serif", color = "#222222",
    box.padding = 0.5, point.padding = 0.3,
    segment.color = "#888888", segment.linewidth = 0.4,
    min.segment.length = 0.2,
    max.overlaps = Inf, seed = 42) +
  coord_sf(xlim = c(-9.5, 4.5), ylim = c(35.5, 44.5), expand = FALSE) +
  labs(title = NULL, x = NULL, y = NULL) +
  theme_AP() +
  theme(
    panel.background = element_rect(fill = "#dce8f0", color = NA),
    panel.grid       = element_line(color = "#cccccc", linewidth = 0.2),
    axis.text        = element_text(size = 8, color = "#555555"),
    plot.margin      = margin(10, 10, 10, 10)
  )
```

```
## Warning in geom_text_repel(data = locations, aes(x = lon, y = lat, label =
## place), : Ignoring unknown parameters: `segment.linewidth`

plot_locations
```



1.2 Load vocabulary data

```
# LOAD VOCABULARY DATA #####

# Irrigators: factors sheet, general list, English translation -----

irrigators <- as.data.table(
  read_xlsx(here("datasets", "input", "carmen_dataset_2.xlsx"), sheet = "factors")
)
irrigators_wc <- irrigators[list == "general" & !is.na(factor_en),
  .(n = .N), by = .(word = factor_en)]
irrigators_wc[, group := "Traditional irrigators"]

# Scientists: Sheet2, general list, cleaned terms -----

scientists <- as.data.table(
  read_xlsx(here("datasets", "input", "scientists_vocabulari.xlsx"), sheet = "Sheet2")
)
scientists_wc <- scientists[list == "general" & !is.na(factor_clean),
  .(n = .N), by = .(word = factor_clean)]
scientists_wc[, group := "Scientists"]

# Top 55 per group -----

irrigators_wc <- setorder(irrigators_wc, -n)[1:60]
scientists_wc <- setorder(scientists_wc, -n)[1:60]

# Combine -----

wc_data <- rbind(irrigators_wc, scientists_wc)
```

1.3 Plot wordcloud

```
# WORDCLOUD #####

plot_wordcloud <- ggplot(wc_data, aes(label = word, size = n)) +
  geom_text_wordcloud(seed = 42, color = "black") +
  scale_size_area(max_size = 4.6) +
  facet_grid(. ~ group) +
  theme_AP()

plot_wordcloud
```


1.4 Calculation of Spain's share

We use the ten global irrigated-area datasets compiled by Puy et al. (2026): GAEZ+2015, GIAM, GMIA, GRIPC, LUH2, Meier, MIRCA 2000, MIRCA OS, Nagaraj, and SPAM2010. For each dataset and spatial resolution (0.2°, 0.4°, 1°), we compute Spain's irrigated area (Mha), total European irrigated area (Mha), and Spain's percentage share.

```
# READ IN DATASETS #####

country_level_irrigated_areas <- fread(here("datasets", "input",
                                             "country_level_irrigated_areas.csv"))

# Calculate -----

country_level_irrigated_areas[continent == "Europe",
                              .(spain_mha = sum(mha[country == "Spain"]),
                                europe_mha = sum(mha),
                                spain_share_pct = 100 * sum(mha[country == "Spain"]) /
                                sum(mha)),
                              by = .(dataset, resolution)]
```

##	dataset	resolution	spain_mha	europe_mha	spain_share_pct
##	<char>	<char>	<num>	<num>	<num>
## 1:	gaez_v4	0.2deg	3.115963	23.84897	13.065402
## 2:	giam	0.2deg	2.702138	38.65570	6.990269
## 3:	gmia	0.2deg	3.231380	11.95049	27.039725
## 4:	gripc	0.2deg	4.003731	23.03500	17.381072
## 5:	luh2	0.2deg	3.135771	13.78261	22.751647
## 6:	meier	0.2deg	4.223571	26.00185	16.243345
## 7:	mirca2000	0.2deg	3.063020	15.32320	19.989427
## 8:	mirca_os	0.2deg	2.872767	10.92628	26.292260
## 9:	nagaraj	0.2deg	4.651206	32.00352	14.533421
## 10:	spam	0.2deg	2.723286	10.92166	24.934715
## 11:	gaez_v4	0.4deg	3.088649	23.62507	13.073608
## 12:	giam	0.4deg	2.686260	38.53872	6.970287
## 13:	gmia	0.4deg	3.192415	11.81225	27.026305
## 14:	gripc	0.4deg	3.966339	23.03867	17.216008
## 15:	luh2	0.4deg	3.123777	13.73316	22.746234
## 16:	meier	0.4deg	4.177631	25.64511	16.290168
## 17:	mirca2000	0.4deg	3.043205	15.29263	19.899813
## 18:	mirca_os	0.4deg	2.840007	10.87587	26.112921
## 19:	nagaraj	0.4deg	4.601698	31.51188	14.603057
## 20:	spam	0.4deg	2.692517	10.80322	24.923284
## 21:	gaez_v4	1deg	2.936647	23.21744	12.648451
## 22:	giam	1deg	2.558272	38.42329	6.658128
## 23:	gmia	1deg	3.056154	11.54011	26.482897
## 24:	gripc	1deg	3.745961	22.80496	16.426079
## 25:	luh2	1deg	2.948508	13.17117	22.386067

## 26:	meier	1deg	3.928146	25.11399	15.641264
## 27:	mirca2000	1deg	2.887533	14.99172	19.260849
## 28:	mirca_os	1deg	2.712319	10.69960	25.349720
## 29:	nagaraj	1deg	4.272359	30.64785	13.940160
## 30:	spam	1deg	2.559707	10.50107	24.375677
##	dataset	resolution	spain_mha	europa_mha	spain_share_pct
##	<char>	<char>	<num>	<num>	<num>

2 References

Puy, Arnald, Olivia Richards, Seth Nathaniel Linga, Samuel Flinders, Carmen Aguiló-Rivera, and Josh Larsen. 2026. “Code and Datasets of Where Irrigation Exists Is Globally Contested.” *Zenodo*, ahead of print, March. <https://doi.org/10.5281/zenodo.19001232>.

3 Session information

```
# SESSION INFORMATION #####
```

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] benchmarkme_1.0.8      cowplot_1.2.0.9000      ggwordcloud_0.6.2
## [4] readxl_1.4.5           here_1.0.2              ggrepel_0.9.6
## [7] rnaturalearthdata_1.0.0 rnaturalearth_1.1.0      sf_1.1-0
## [10] ggplot2_4.0.1.9000     data.table_1.18.2.1     sensobol_1.1.6
##
## loaded via a namespace (and not attached):
## [1] benchmarkmeData_1.0.4 gtable_0.3.6            xfun_0.55
## [4] lattice_0.22-7        vctrs_0.7.1             tools_4.5.2
## [7] Rdpack_2.6.4          generics_0.1.4          parallel_4.5.2
## [10] tibble_3.3.1          proxy_0.4-29            pkgconfig_2.0.3
## [13] Matrix_1.7-4          KernSmooth_2.23-26      RColorBrewer_1.1-3
## [16] S7_0.2.1              lifecycle_1.0.5         stringr_1.6.0
## [19] compiler_4.5.2        farver_2.1.2            tinytex_0.58
## [22] codetools_0.2-20      litedown_0.7            htmltools_0.5.9
## [25] class_7.3-23          yaml_2.3.12             pillar_1.11.1
## [28] classInt_0.4-11       iterators_1.0.14        foreach_1.5.2
## [31] commonmark_2.0.0      tidysselect_1.2.1       digest_0.6.39
## [34] stringi_1.8.7         dplyr_1.1.4            labeling_0.4.3
## [37] rprojroot_2.1.1       fastmap_1.2.0           grid_4.5.2
## [40] colorspace_2.1-2      cli_3.6.5              magrittr_2.0.4
## [43] e1071_1.7-17          withr_3.0.2            scales_1.4.0
## [46] rmarkdown_2.30        httr_1.4.8             otel_0.2.0
## [49] cellranger_1.1.0      png_0.1-8              evaluate_1.0.5
## [52] knitr_1.51            rbibutils_2.4          doParallel_1.0.17
```

```
## [55] markdown_2.0      rlang_1.1.7      gridtext_0.1.5
## [58] Rcpp_1.1.1         glue_1.8.0       DBI_1.3.0
## [61] xml2_1.5.2         rstudioapi_0.17.1 R6_2.6.1
## [64] units_1.0-0
```

```
## Return the machine CPU -----
```

```
cat("Machine:    "); print(get_cpu()$model_name)
```

```
## Machine:
```

```
## [1] "Apple M1 Max"
```

```
## Return number of true cores -----
```

```
cat("Num cores:   "); print(parallel::detectCores(logical = FALSE))
```

```
## Num cores:
```

```
## [1] 10
```

```
## Return number of threads -----
```

```
cat("Num threads: "); print(parallel::detectCores(logical = FALSE))
```

```
## Num threads:
```

```
## [1] 10
```